22CH203 - ENGINEERING CHEMISTRY II

UNIT III & LP 3 – METAL CORROSION AND ITS PREVENTION

1. FACTORS INFLUENCING RATE OF CORROSION

The rate and extent of corrosion mainly depends on

1. Nature of the metal.
2. Nature of the environment.

1.1. Nature of the Metal

1.1.1. Position in EMF series

The extent of corrosion depends on the position of the metal in the EMF series. Metals above the hydrogen in the EMF series get corroded vigorously. The lower the reduction potential, the greater the rate of corrosion. When two metals are in electrical contact, the more active metal (or the metal having a high negative reduction potential) undergoes corrosion.

The rate and severity of corrosion depend on the difference in their positions in the EMF series. The greater the difference faster the corrosion rate.

1.1.2. Relative areas of the anode and cathode

The rate of corrosion will be higher when the cathodic area is larger. When the cathodic area is larger, the demand for electrons will be greater and this results in an increased rate of corrosion (dissolution) of metals at the anodic area.

1.1.3. Purity of the metal

The 100% pure metal will not undergo any type of corrosion. However, the presence of impurities in a metal creates heterogeneity and thus galvanic cells are set up with distinct anodic and cathodic areas in the metal. The higher the percentage of impurity, the faster the rate of corrosion of the anodic metal. The effect of impurities on the rate of corrosion of zinc is given below.

%Purity of Zn	99.9999	99.99	99.95
Corrosion rate	1	2650	5000

1.1.4. Over voltage or overpotential

The overvoltage of a metal in a corrosive environment is inversely proportional to the corrosive rate. Example: The normal hydrogen overvoltage of zinc metal, when it is dipped in 1 M H_2SO_4 is 0.7 volts. Here the rate of corrosion is low. By adding a small amount of impurity like $CuSO_4$ to H_2SO_4 the hydrogen overvoltage is reduced to 0.33 V. This results in an increased rate of corrosion of zinc metal.

1.1.5. Nature of the surface film

The nature of the oxide film formed on the metal surface decides the extent of corrosion which can be decided by the pilling-Bedworth rule:

In the case of alkali and alkaline earth metals such as Mg, Ca, etc form oxide, whose volume is less than the volume of the metal. Hence the oxide film will be porous and non-protective and bring about further corrosion. But heavy metals like Al, Cr etc form oxide, whose volume is greater than that of the metal. Hence the oxide film will be non-porous and protective and prevent further corrosion.

1.1.6. Nature of the corrosion product

If the corrosion product is soluble in the corroding medium, the corrosion rate will be faster. Similarly, if the corrosion product is volatile (like MoO_3 on the Mo surface), the corrosion rate will be faster.

1.2. Nature of Environment

1.2.1. Temperature

The rate of corrosion is directly proportional to temperature. This is because the rate of chemical reaction and the rate of diffusion of the ions increases with rise in temperature. Hence the rate of corrosion increases with temperature.

1.2.2. Humidity

The rate of corrosion will be higher when the humidity in the environment is high. The moisture acts as a solvent for the oxygen in the air to produce the electrolyte, which is essential for setting up a corrosion cell.

1.2.3. Presence of corrosive gases

The acidic gases like CO_2, SO_2, H_2S and fumes of HCl, H_2SO_4, etc produce electrolytes that are acidic and increase the electrochemical corrosion.

1.2.4. Presence of suspended particles

Particles like NaCl, $(NH_4)_2SO_4$ along with moisture act as powerful electrolytes and thus accelerate the electrochemical corrosion.

1.2.5. Effect of pH

The possibility of corrosion with respect to the pH of the electrolytic solution and the electrode potential of the metal is correlated with the help of a Pourbaix diagram. The Pourbaix diagram for iron in water is shown in Fig. 3.1.

It shows clearly the zones of corrosion, immunity and passivity in the diagram. ‘Z’ is the point where pH =6 and the electrode potential = -0.4 V. It is present in the corrosion Zone. This clearly shows that iron rusts in water under those conditions. In actual practice, it is observed to be true. From the diagram (Fig. 3.1) it is clear that the rate of corrosion can be altered by shifting the point 'Z' into immunity or passivity regions.

The iron will be immune to corrosion if the potential is changed to about - 0.8 V by applying an external current. On the other hand, the rate of corrosion of iron can also be reduced by moving into the passivity region by applying positive potential.

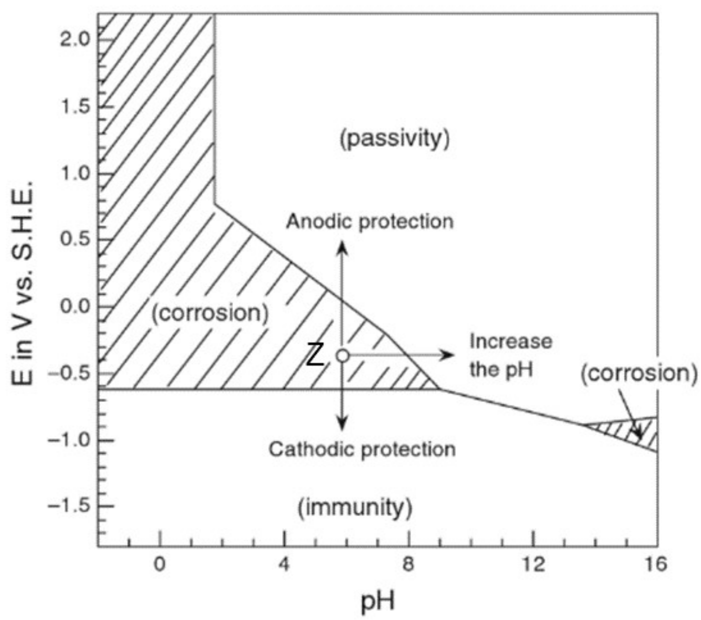

Fig. 3.1. Simplified Pourbaix diagram for Iron at 25 °C.

The diagram indicates that the rate of corrosion can also be reduced by increasing the pH of the solution by adding alkali. Thus, the rate of corrosion will be maximum when the corrosive environment is acidic, i.e. pH is less than 7.

Discussion questions:

1. How does the position of a metal in the EMF series influence its susceptibility to corrosion?
2. Explain the relationship between the purity of a metal and its rate of corrosion.
3. Analyze the role of humidity and corrosive gases in accelerating the electrochemical corrosion process.
4. Describe the factors influencing the nature of the surface film formed on metal surfaces and their role in corrosion prevention.